

1. What is SQL?

SQL (Structured Query Language) is a standard language used to store, retrieve, and manipulate data in relational databases.

2. What is the difference between INNER JOIN and LEFT JOIN?

- INNER JOIN returns only matching rows from both tables.
- LEFT JOIN returns all rows from the left table, and matched rows from the right table (or NULLs if no match).

3. What is the difference between ON and WHERE clause in SQL joins?

- ON is used to define the join condition between two tables.
- WHERE is used to filter the final result set after the join is performed.

4. What is the difference between UNION and UNION ALL?

- UNION removes duplicate rows.
- UNION ALL keeps all rows, including duplicates.

5. What is a Primary Key?

A Primary Key uniquely identifies each record in a table and does not allow NULLs or duplicates.

6. What is a Foreign Key?

A Foreign Key is a column that creates a relationship between two tables by referencing the primary key of another table.

7. What is a Subquery?

A Subquery is a query within another SQL query. It can be used in SELECT, INSERT, UPDATE, or DELETE statements.

8. What is Normalization?

Normalization is the process of organizing data to reduce redundancy and improve data integrity.

9. What is a View in SQL?

A View is a virtual table based on the result set of an SQL query. It does not store data physically.

10. What is the difference between DELETE and TRUNCATE?

- DELETE removes specific rows and can be rolled back.
- TRUNCATE removes all rows and cannot be rolled back (in most databases).